How to Write a Survey Paper:

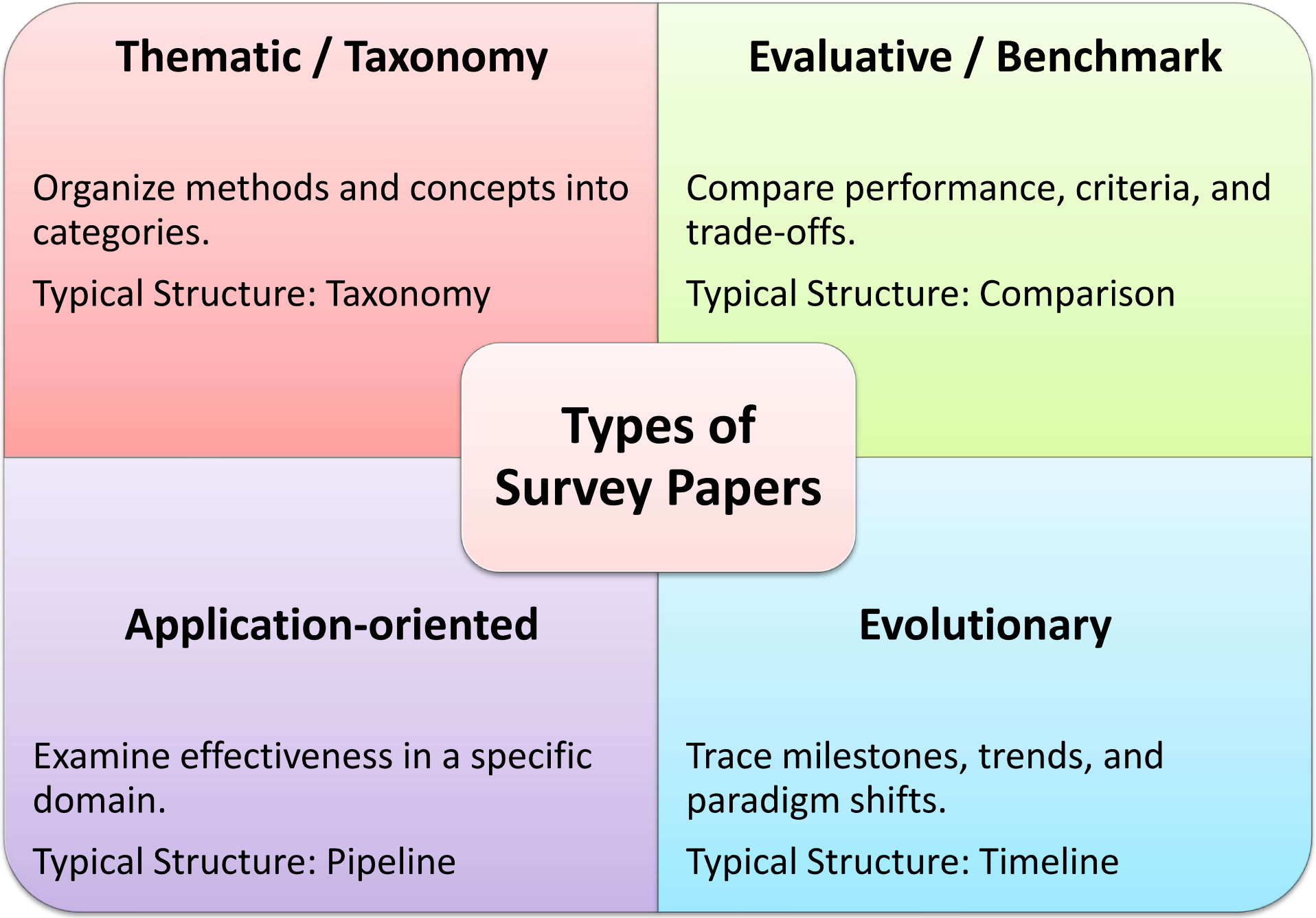
From Literature Search to Defensible Research Gaps

What is a survey paper?

- A **survey paper** reviews existing research on a specific topic in order to:
  - ✓ Understand what has been done.
  - ✓ Compare methods and findings.
  - ✓ Identify limitations, research gaps, and future directions.

- The goal is to understand the research field as a whole, not just summarize individual papers.

- A survey may be **narrative**, **critical**, **systematic**, or a ***combination*** of these approaches.



* These approaches may overlap; a survey can combine more than one.

Traditional Survey vs. Systematic Literature Review

Feature	Traditional Survey	Systematic Literature Review
Primary goal	Understand the field, compare approaches, identify challenges	Answer focused research questions
Search strategy	May be selective; documentation varies	Uses a documented search strategy
Paper selection	Guided by relevance and expert judgment	Applies predefined eligibility criteria
Analysis and synthesis	Critically compares methods, themes, and limitations	Synthesizes evidence to answer the review questions
Quality assessment	May assess study quality formally	May apply predefined quality criteria
Transparency	Depends on documentation of review decisions	Explicitly reports review procedures

PRISMA 2020

- **P**referred **R**eporting **I**tems for **S**ystematic Reviews and **M**eta-**A**nalyses.

- A widely used **reporting guideline** for systematic reviews.

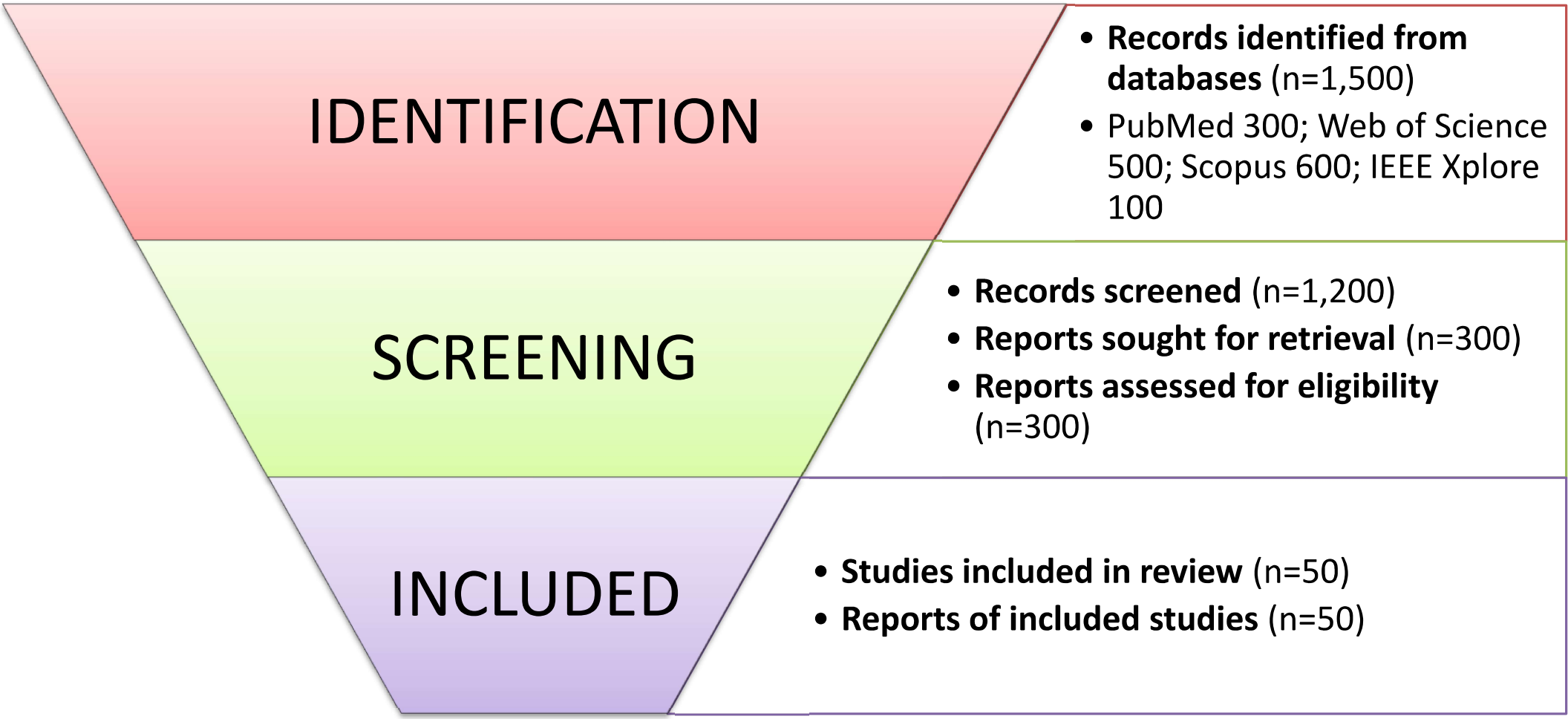
- Promotes **transparent, complete, and clear reporting** of the review process.

- **PRISMA guides reporting, not how to conduct the research.**

https://www.prisma-statement.org/prisma-2020

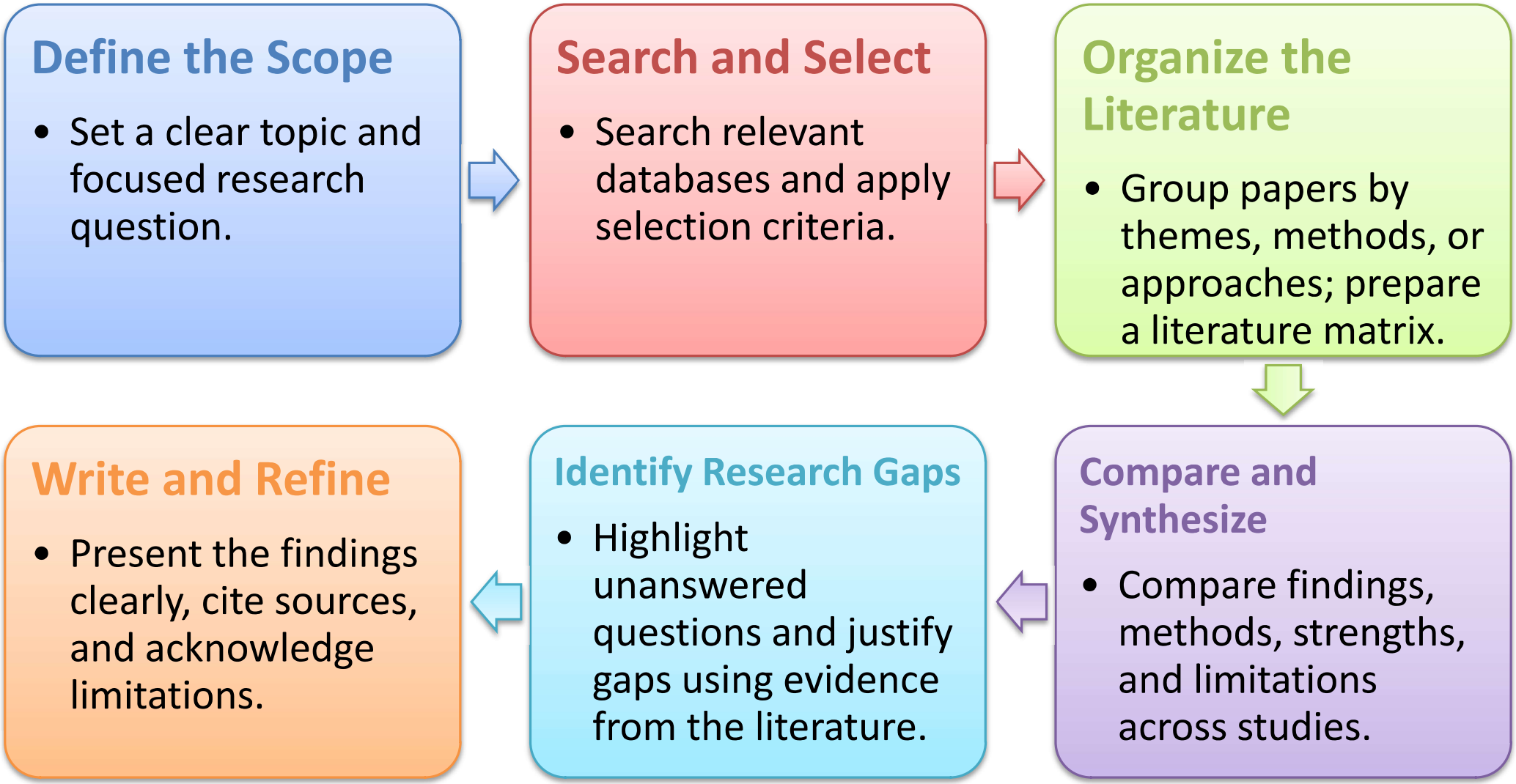
Key Components of PRISMA 2020

- 27-item checklist
- A **3-step funnel** that visually maps out a literature search:



**** Numbers are illustrative.***

Practical Steps for Writing a Survey Paper



Suggestions: Plan and Search

- **Define a focused scope.**
  - Write 2–4 guiding questions before searching.
- **Choose the survey type early.**
  - Match taxonomy, benchmark, application, or evolutionary to your goal.
- **Check existing surveys.**
  - Confirm your paper adds something new.
- **Fix inclusion criteria.**
  - Set year range, venues, and quality filters up front.

- **Search several databases.**
  - Use Scopus, Web of Science, IEEE Xplore, and Google Scholar.
- **Log every search.**
  - Save strings, filters, dates, and result counts.
- **Follow citation trails.**
  - Check references and citing papers of key works.
- **Use a reference manager.**
  - Zotero or Mendeley keeps sources organized and de-duplicated.

Suggestions: Synthesize and Write

- **Build a comparison matrix.**
  - Track method, data, metrics, results, and limitations per paper.
- **Create your own taxonomy.**
  - Group by ideas, not by author or year.
- **Use tables and figures.**
  - Visuals should compare and summarize, not decorate.
- **Support every gap with evidence.**
  - Cite the papers that show what is missing.

- **Outline before drafting.**
  - Let the taxonomy or research questions shape the sections.
- **Write critically.**
  - Explain why methods differ, not just what they do.
- **Verify every citation.**
  - Read the source; do not rely on AI summaries.
- **Get feedback, then target a journal.**
  - Ask a peer to review, and follow the journal's guidelines.

Dos and Don'ts

- ✓ **Do look for patterns, not just papers.**
  - Connect findings to reveal broader insights.
- ✓ **Do question the evidence.**
  - Examine assumptions, data, and metrics.
- ✓ **Do identify gaps through synthesis.**
  - Uncover contradictions and unresolved questions.
- ✓ **Do offer a perspective.**
  - Develop a useful framework or interpretation.

- × **Don't make the scope too broad.**
  - Keep the focus manageable.
- × **Don't just summarize papers.**
  - Synthesize findings and compare evidence.
- × **Don't assume research gaps.**
  - Validate them against the existing literature.
- × **Don't prioritize quantity, novelty, or AI-generated summaries over verified quality.**
  - Focus on meaningful insights and verified evidence.

Stay curious. Keep questioning.

Happy Research!